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CHAPTER-1

INTRODUCTION

Our aim is to create a user-friendly online e-canteen platform—experienced using HTML, CSS, and JavaScript. This website aims to simplify college canteen services for students and staff. It allows users to order food online, track orders in real-time, and access various services from a single interface. The website offers features such as digital menu browsing, cart management, cashless payments, inventory tracking, and even feedback submission. This system enhances convenience for students and streamlines operations for canteen staff.

1.1 Problem Definition

Most existing college canteen systems rely on manual processes, leading to long queues, billing errors, cash handling risks, and poor inventory control. Key functions like real-time order tracking, dynamic menu updates, personalized recommendations, and demand forecasting are often missing. This limits student satisfaction and efficient canteen operations. Our project aims to overcome these gaps by offering a smarter, more user-focused e-canteen system.

1.2 Existing System

- Offers only basic features like manual ordering, cash payments, and simple stock checks.
- Lacks personalized services like order customization and student preferences.
- No real-time queue status or order delivery tracking.
- Limited multi-payment options and no analytics for sales trends.

1.3 Proposed System

- Provides personalized services such as menu customization and favorite order history.
- Includes real-time order tracking and kitchen notifications for faster service.
- Supports multiple payment gateways like UPI, cards, and campus wallets.
- Offers detailed analytics, sales reports, and feedback tools.
- Integrated with college portals and real-time inventory alerts.

1.4 Literature Review

Many existing canteen management systems focus on basic tasks like ordering and billing. However, studies show that modern campus users expect real-time features and personalization. There is a growing need for systems that offer multi-payment support, order customization, live tracking, and better operational insights. Our project aims to improve on these gaps by combining smart features into a user-friendly and efficient college e-canteen system

CHAPTER-2

SYSTEM REQUIREMENTS

2.1 Hardware & Software Requirements

Hardware Requirements:

- Processor: Intel i3 or above
- RAM: Minimum 4 GB
- Storage: 500 GB HDD or 256 GB SSD
- Display: 13” or larger monitor with 1024x768 resolution
- Internet Connection: Required for online ordering and payment gateways

Software Requirements:

- Operating System: Windows 10/Linux/Mac
- Frontend: HTML, CSS, JavaScript
- Backend: PHP / Node.js
- Database: MySQL

2.2 Software Requirements Specification (SRS)

The SRS document outlines the functional and non-functional requirements of the College E-Canteen System:

Functional Requirements (FR):

- User registration and login with college email verification
- Digital menu browsing and search functionality
- Online food ordering with cart management
- Cashless payment integration (UPI, cards, campus wallet)

Non-Functional Requirements (NFR):

- Fast and responsive system (load time < 3 seconds)
- Easy to use (intuitive student-friendly interface)
- High security with role-based access and transaction encryption
- 99.9% system availability during peak hours

CHAPTER-3

SYSTEM DESIGN

3.1 Modules of Systems

- **User Module:** This module handles user registration and login using college email verification. It allows students and staff to browse the digital menu, search for food items, and manage their shopping cart.
- **Order & Tracking Module:** This core module facilitates online food ordering. It provides real-time order tracking and sends kitchen notifications to ensure faster service and transparency.
- **Payment Module:** This module manages cashless transactions, supporting multiple gateways such as UPI, credit/debit cards, and dedicated campus wallets.
- **Admin Module:** Designed for canteen staff, this module features a dashboard for dynamic menu updates, real-time inventory alerts, and demand forecasting.
- **Analytics & Feedback Module:** This module collects student ratings and feedback. It also provides detailed sales reports and analytics to help the canteen staff track trends and reduce waste.

3.2 UML Diagrams

- **Use Case Diagram:** This diagram illustrates the interactions between the primary actors—Students and Canteen Staff—and the system. It maps out functionalities like digital menu browsing, order placement, payment processing, and inventory management.
- **Class Diagram:** This describes the underlying structure of the system by showing the relationship between various entities such as User, FoodItem, Order, Payment, and Inventory.
- **Activity Diagram:** This diagram displays the workflow of the ordering process, starting from the user login, through menu selection and cart management, to the final step of cashless payment and order confirmation.
- **Sequence Diagram:** This shows how different objects in the e-canteen system interact over time, such as the communication between the user interface, the payment gateway, and the kitchen notification system when an order is placed.

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IMPLEMENTATION

4.1 Sample Code

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Smart E-Canteen</title>

<link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600&display=swap"
rel="stylesheet">

<style>

*{

    margin:0;

    padding:0;

    box-sizing:border-box;

    font-family:'Poppins',sans-serif;

}

body{

    background:linear-gradient(135deg,#fdfbfb,#ebedee);

}

header{

    background:#ff6b35;
```

```
    color:white;
    padding:18px;
    text-align:center;
    font-size:26px;
    font-weight:600;
}

nav{
    background:#222;
    display:flex;
    justify-content:center;
    gap:25px;
    padding:12px;
    flex-wrap:wrap;
}

nav a{
    color:white;
    text-decoration:none;
    cursor:pointer;
    font-weight:500;
}

nav a:hover{
    color:#ff6b35;
}
```

```
.section{
  display:none;
  padding:30px;
}

.active{
  display:block;
}

.grid{
  display:grid;
  grid-template-columns:repeat(auto-fit,minmax(250px,1fr));
  gap:25px;
}

.card{
  background:white;
  border-radius:15px;
  overflow:hidden;
  box-shadow:0 8px 20px rgba(0,0,0,0.1);
  transition:0.3s;
}

.card:hover{
  transform:translateY(-6px);
}
```

```
.card img{
    width:100%;
    height:200px;
    object-fit:cover;
}
.card-content{
    padding:15px;
}
.price{
    color:#ff6b35;
    font-weight:bold;
    margin:8px 0;
}
button{
    background:#ff6b35;
    border:none;
    padding:8px 15px;
    color:white;
    border-radius:8px;
    cursor:pointer;
}
button:hover{
    background:#e85a2a;
}

.cart-item{
    background:white;
```

```

padding:12px;
margin:10px 0;
border-radius:10px;
box-shadow:0 5px 10px rgba(0,0,0,0.1);
}

footer{
background:#222;
color:white;
text-align:center;
padding:15px;
margin-top:40px;
}
</style>
</head>
<body>
<header><img alt="Smart E-Canteen logo" data-bbox="193 578 208 596"/> Smart E-Canteen</header>
<nav>
<a onclick="showSection('home')">Home</a>
<a onclick="showSection('breakfast')">Breakfast</a>
<a onclick="showSection('lunch')">Lunch</a>
<a onclick="showSection('snacks')">Snacks</a>
<a onclick="showSection('drinks')">Drinks</a>
<a onclick="showSection('cart')">Cart</a>
</nav>
<!-- HOME -->
<div id="home" class="section active">

```

```

<h2>Welcome to Smart E-Canteen</h2>
<p>Order your favorite meals easily. Fresh food, fast delivery & cashless payments.</p>

</div>
<!-- BREAKFAST -->
<div id="breakfast" class="section">
<h2>Breakfast Menu</h2>
<div class="grid">
<div class="card">

<div class="card-content">
<h3>Masala Dosa</h3>
<p class="price">₹50</p>
<button onclick="addToCart('Masala Dosa',50)">Add to Cart</button>
</div>
</div>
<div class="card">

<div class="card-content">
<h3>Idli Sambar</h3>
<p class="price">₹40</p>
<button onclick="addToCart('Idli Sambar',40)">Add to Cart</button>
</div>
</div>
<div class="card">


```

```

<div class="card-content">
<h3>Poori Bhaji</h3>
<p class="price">₹45</p>
<button onclick="addToCart('Poori Bhaji',45)">Add to Cart</button>
</div>
</div>
</div>
</div>
<!-- LUNCH -->
<div id="lunch" class="section">
<h2>Lunch Menu</h2>
<div class="grid">
<div class="card">

<div class="card-content">
<h3>Veg Thali</h3>
<p class="price">₹90</p>
<button onclick="addToCart('Veg Thali',90)">Add to Cart</button>
</div>
</div>
<div class="card">

<div class="card-content">
<h3>Biriyani</h3>
<p class="price">₹120</p>
<button onclick="addToCart('Biriyani',120)">Add to Cart</button>
</div>

```

```

</div>

<div class="card">



<div class="card-content">

<h3>Fried Rice</h3>

<p class="price">₹80</p>

<button onclick="addToCart('Fried Rice',80)">Add to Cart</button>

</div>

</div>

</div>

</div>

<!-- SNACKS -->

<div id="snacks" class="section">

<h2>Snacks</h2>

<div class="grid">

<div class="card">



<div class="card-content">

<h3>Samosa</h3>

<p class="price">₹20</p>

<button onclick="addToCart('Samosa',20)">Add to Cart</button>

</div>

</div>

<div class="card">



<div class="card-content">

```

```

<h3>Burger</h3>
<p class="price">₹60</p>
<button onclick="addToCart('Burger',60)">Add to Cart</button>
</div>
</div>

<div class="card">

<div class="card-content">
<h3>Sandwich</h3>
<p class="price">₹50</p>
<button onclick="addToCart('Sandwich',50)">Add to Cart</button>
</div>
</div>
</div>
</div>
<!-- DRINKS -->
<div id="drinks" class="section">
<h2>Drinks</h2>
<div class="grid">
<div class="card">

<div class="card-content">
<h3>Cold Coffee</h3>
<p class="price">₹60</p>
<button onclick="addToCart('Cold Coffee',60)">Add to Cart</button>
</div>

```

```

</div>

<div class="card">
  
  <div class="card-content">
    <h3>Fresh Lime</h3>
    <p class="price">₹30</p>
    <button onclick="addToCart('Fresh Lime',30)">Add to Cart</button>
  </div>
</div>

<div class="card">
  
  <div class="card-content">
    <h3>Mango Shake</h3>
    <p class="price">₹70</p>
    <button onclick="addToCart('Mango Shake',70)">Add to Cart</button>
  </div>
</div>
</div>
</div>
</div>

<!-- CART -->
<div id="cart" class="section">
  <h2>Your Cart</h2>
  <div id="cart-items"></div>
  <h3 id="total"></h3>
  <button onclick="clearCart()">Clear Cart</button>
</div>

<footer>© 2026 Smart E-Canteen</footer>

```

```

<script>
function showSection(id){
    document.querySelectorAll('.section').forEach(sec=>sec.classList.remove('active'));
    document.getElementById(id).classList.add('active');
    if(id==='cart') loadCart();
}

function addToCart(name,price){
    let cart=JSON.parse(localStorage.getItem('cart'))||[];
    cart.push({name,price});
    localStorage.setItem('cart',JSON.stringify(cart));
    alert(name+" added to cart!");
}

function loadCart(){
    let cart=JSON.parse(localStorage.getItem('cart'))||[];
    let container=document.getElementById('cart-items');
    let total=0;
    container.innerHTML="";
    cart.forEach(item=>{
        total+=item.price;
        container.innerHTML+=`
        <div class="cart-item">
        <h4>${item.name}</h4>
        <p>₹${item.price}</p>
        </div>`;
    });
}

```

```
        document.getElementById('total').innerText="Total: ₹"+total;
    }
    function clearCart(){
        localStorage.removeItem('cart');
        loadCart();
    }
</script>
</body>
</html>
```

EXPLANATION:

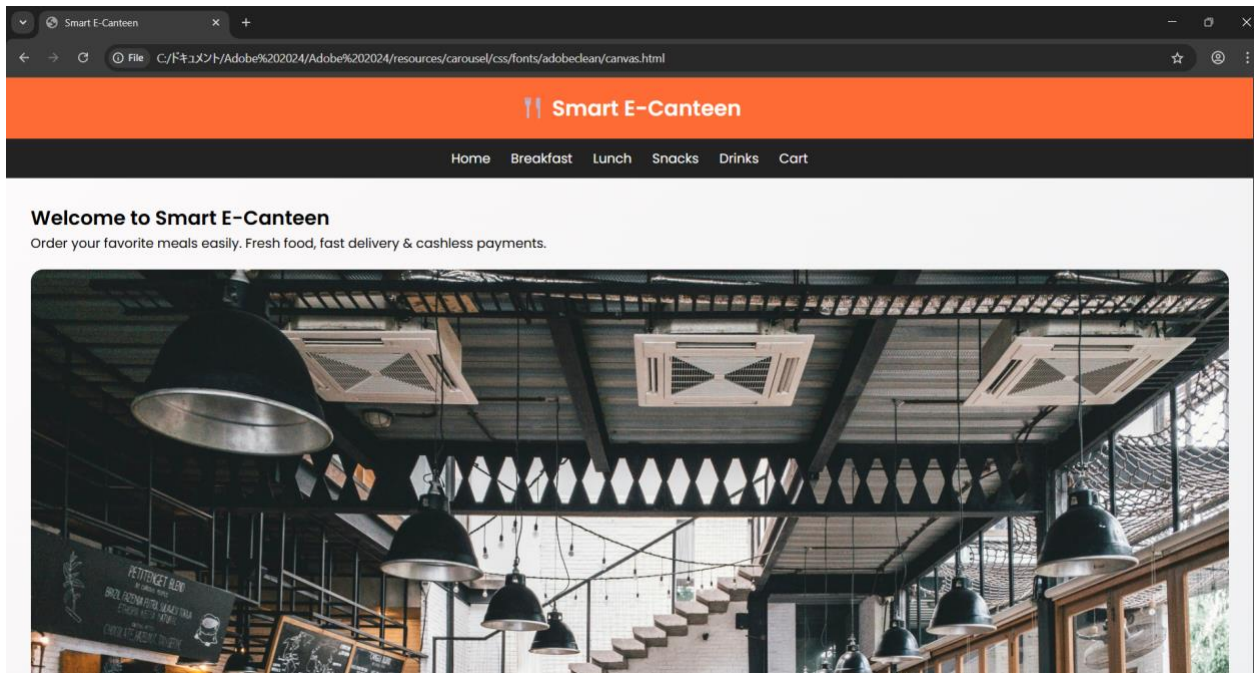
- It is a **single-page E-Canteen website** built using HTML, CSS, and JavaScript.
- The navigation bar switches between sections (Home, Breakfast, Lunch, Snacks, Drinks, Cart) without reloading the page.
- Food items are displayed in card format with image, name, price, and “Add to Cart” button.
- When a user clicks “Add to Cart,” the item is stored in **LocalStorage**.
- The Cart section shows selected items, calculates the total amount, and allows clearing the cart.
- The design is responsive and user-friendly for smooth browsing.

CHAPTER-5

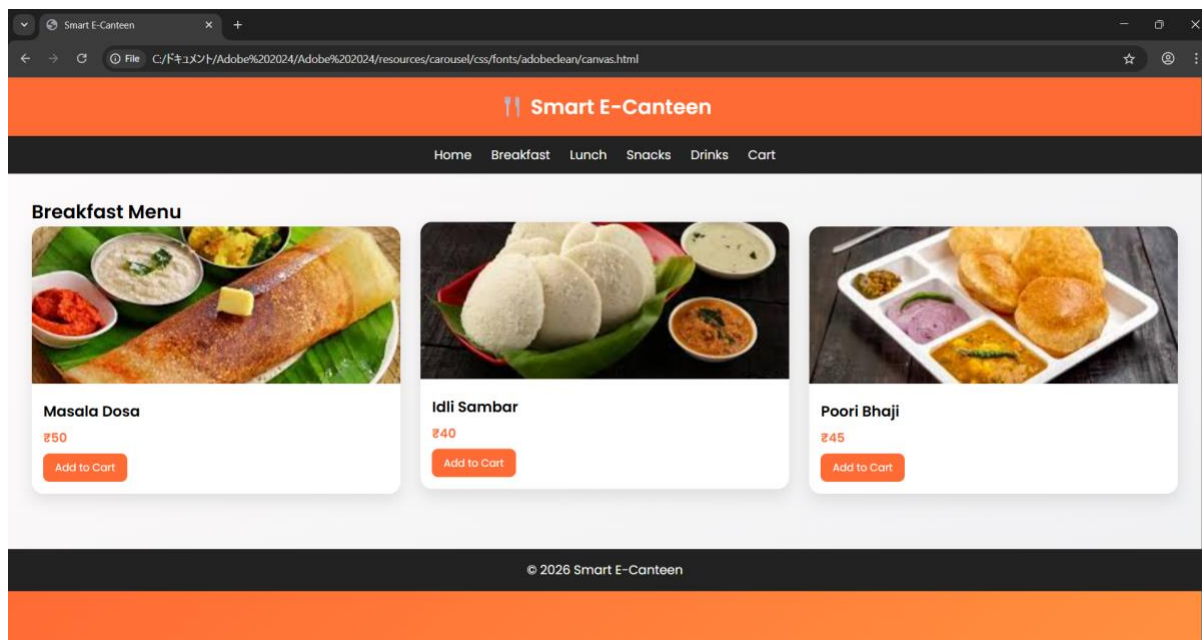
RESULTS

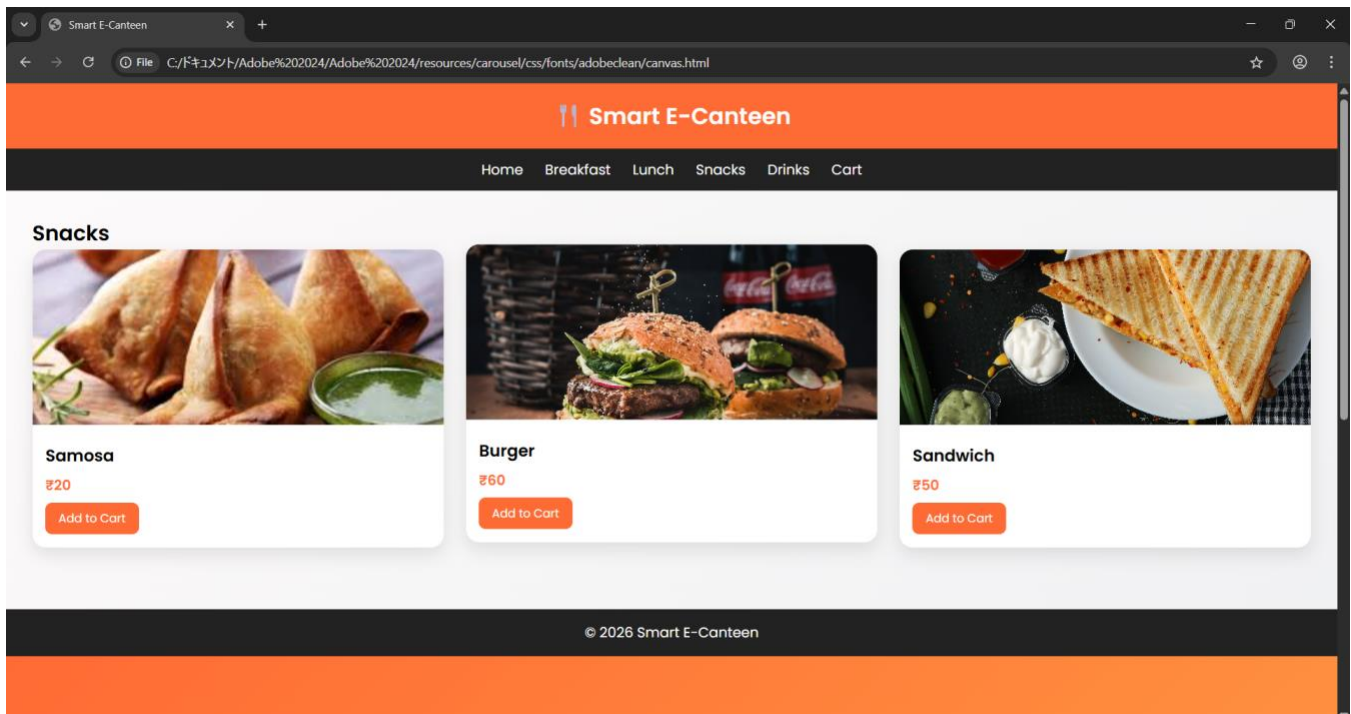
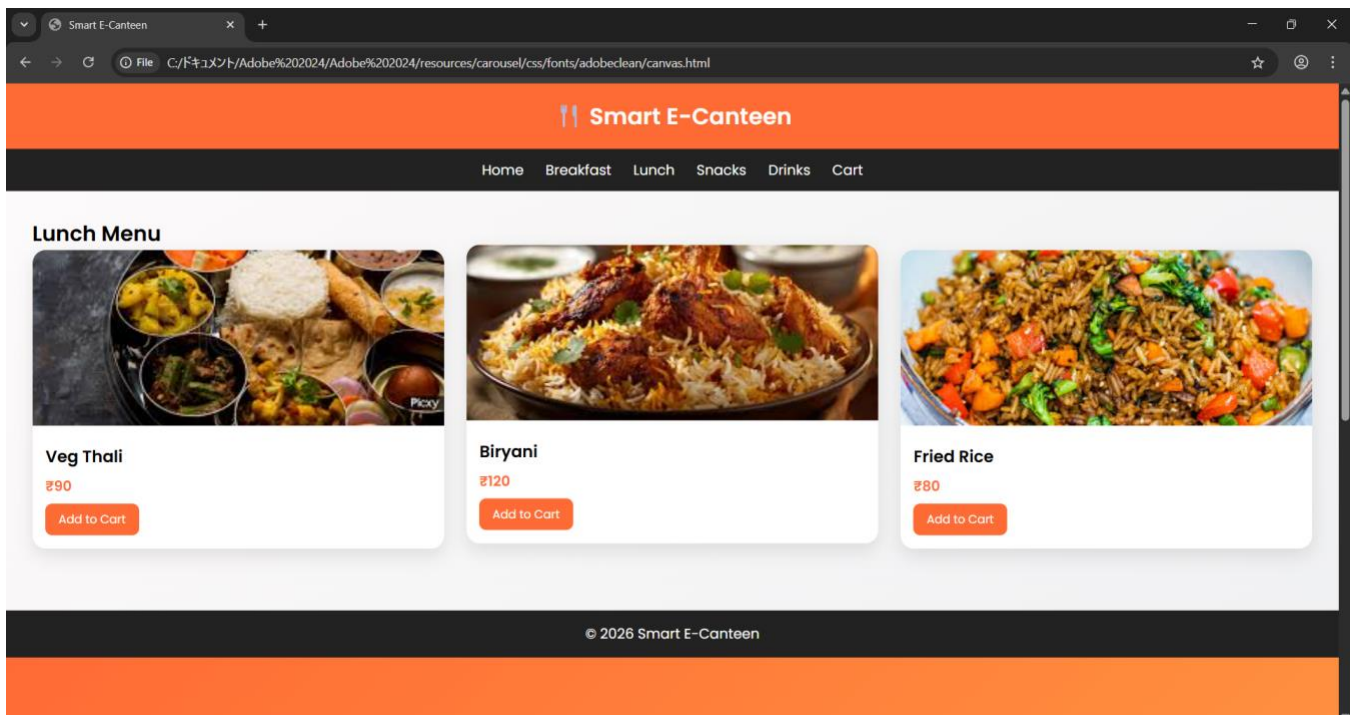
5.1 Output Screens

HOME

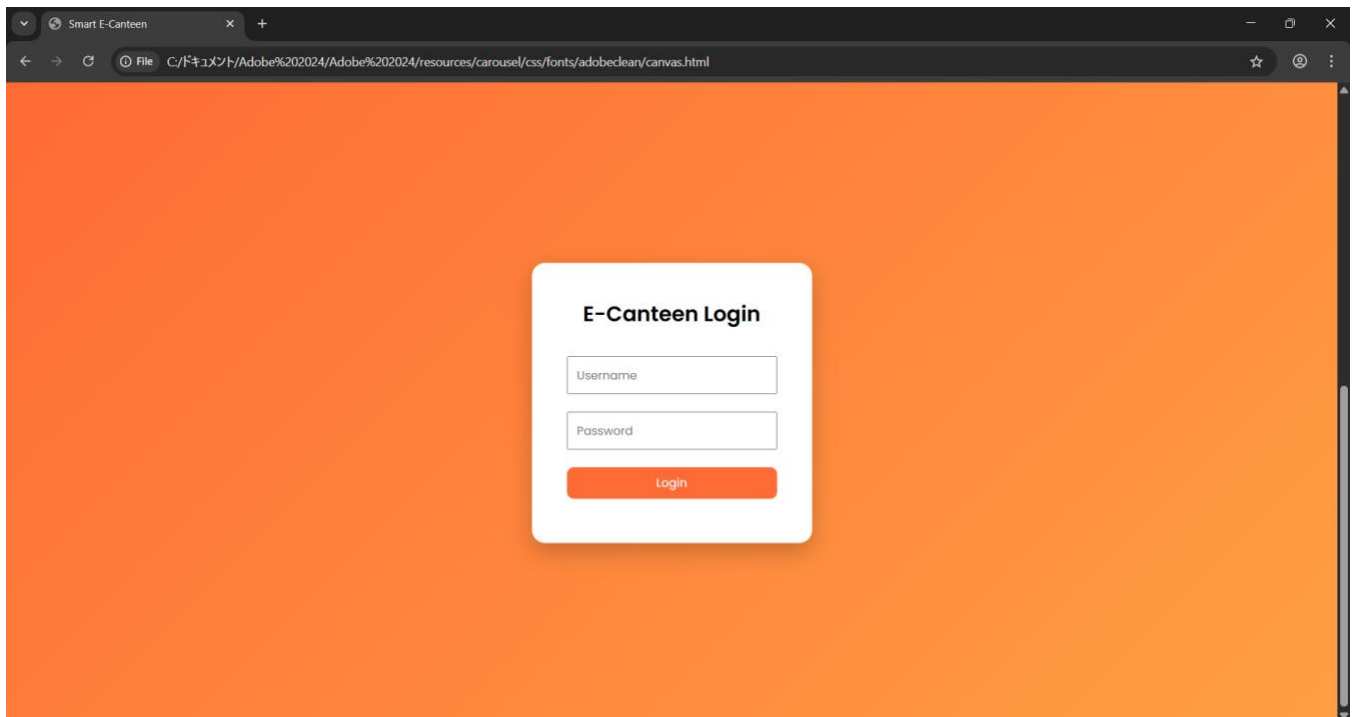


MENU

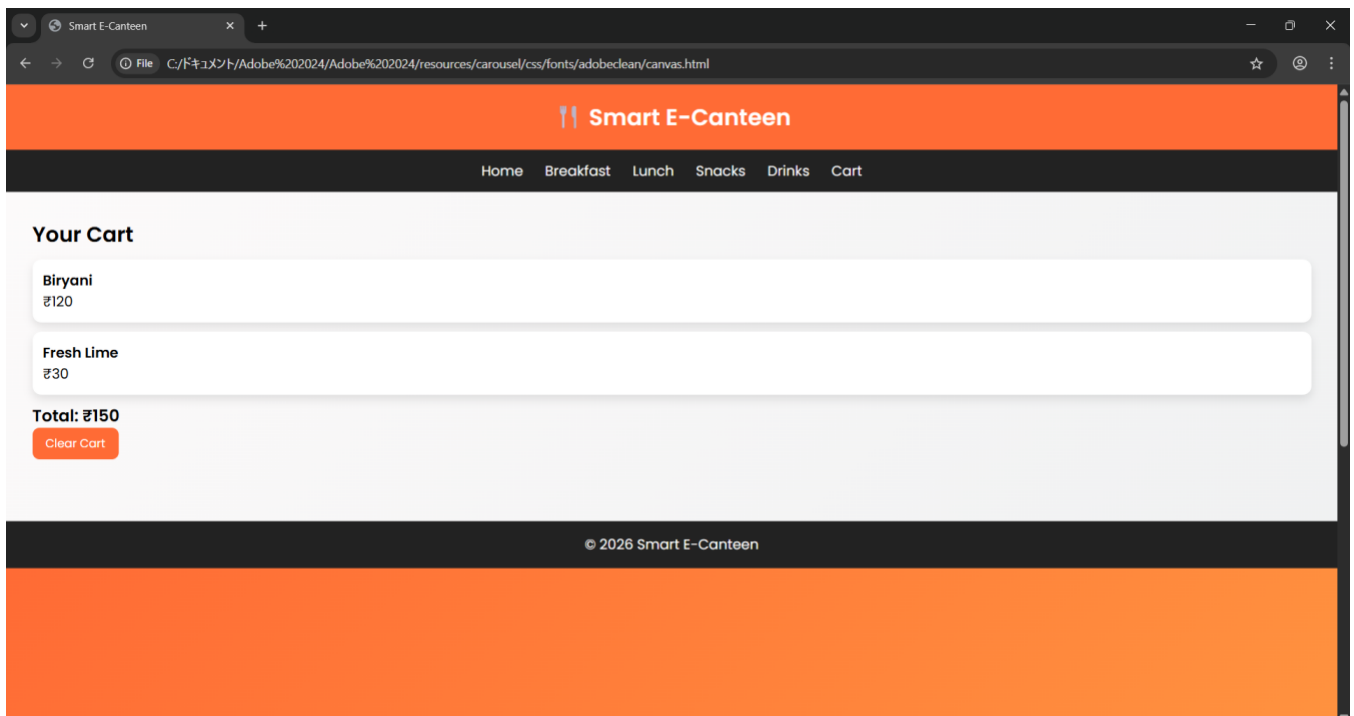




LOGIN



CART



CHAPTER-6

CONCLUSION

- The Smart E-Canteen system successfully provides a user-friendly and efficient digital solution for college canteen management. It simplifies food ordering by allowing users to browse menus, add items to cart, and place orders online without standing in queues. The system enhances convenience, reduces manual errors, and supports cashless transactions.
- By integrating features like real-time menu display, cart management, login functionality, and responsive design, the proposed system improves both student experience and canteen operations. It is scalable, secure, and adaptable for future enhancements such as online payments, inventory tracking, and admin dashboards.
- Overall, the Smart E-Canteen platform modernizes traditional canteen services and contributes to a smarter, technology-driven campus environment

CHAPTER-7

REFERENCES

- W3Schools. *HTML, CSS, and JavaScript Tutorials*. Available at: <https://www.w3schools.com>
- Mozilla Developer Network (MDN). *Web Development Documentation*. Available at: <https://developer.mozilla.org>
- Unsplash. *Free Food Images for Website Design*. Available at: <https://unsplash.com>
- Oracle Corporation. *MySQL Documentation*. Available at: <https://dev.mysql.com/doc>
- PHP Official Documentation. *PHP Manual*. Available at: <https://www.php.net/docs.php>
- Node.js Foundation. *Node.js Documentation*. Available at: <https://nodejs.org>
- Pressman, R. S. (2010). *Software Engineering: A Practitioner's Approach*. McGraw-Hill Education.
- • Sommerville, I. (2016). *Software Engineering (10th Edition)*. Pearson Education.
- • Elmasri, R., & Navathe, S. B. (2015). *Fundamentals of Database Systems*. Pearson.
- • Stallings, W. (2018). *Cryptography and Network Security*. Pearson Education.
- • WHATWG. *HTML Living Standard*. Available at: <https://html.spec.whatwg.org>
- • CSS Working Group. *CSS Specifications*. Available at: <https://www.w3.org/Style/CSS>
- • ECMA International. *ECMAScript Language Specification*. Available at: <https://www.ecma-international.org>
- • Google Developers. *Web Performance Best Practices*. Available at: <https://developers.google.com/web>
- • Bootstrap Documentation. *Responsive Web Design Framework*. Available at: <https://getbootstrap.com>
- • XAMPP Official Documentation. *Apache, MySQL Setup Guide*. Available at: <https://www.apachefriends.org>
- • OWASP Foundation. *Web Application Security Guidelines*. Available at: <https://owasp.org>